ECET 250E-Linear Circuit 1
Problemset 3-Chapter 3- Methods of Analysis

Name: _____ Marks: _____

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1. Chapter 3, Problem 1.

Given $\mathbf{R_1 = 4\ k\Omega}$, $\mathbf{R_2 = 2\ k\Omega}$, and $\mathbf{R_3 = 2\ k\Omega}$, determine the value of $\mathbf{I_x}$ using nodal analysis.

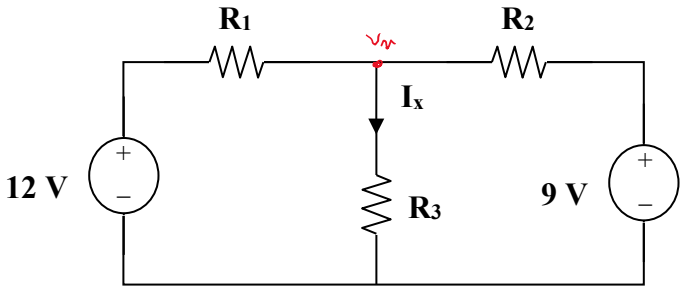


Figure 3.50
For Prob. 3.1

2. Chapter 3, Problem 2.

For the circuit in Fig. 3.51, obtain v_1 and v_2 using nodal analysis.

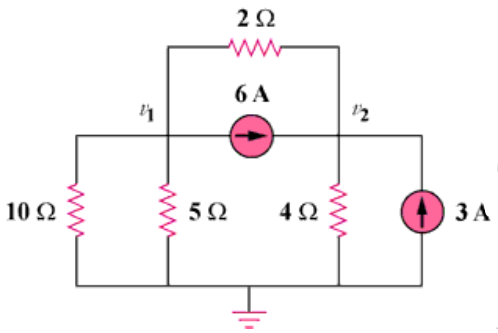


Figure 3.51
For Prob. 3.2.

3. Chapter 3, Problem 3.

Find the currents i_1 through i_4 and the voltage v_o in the circuit in Fig. 3.52 using nodal analysis.

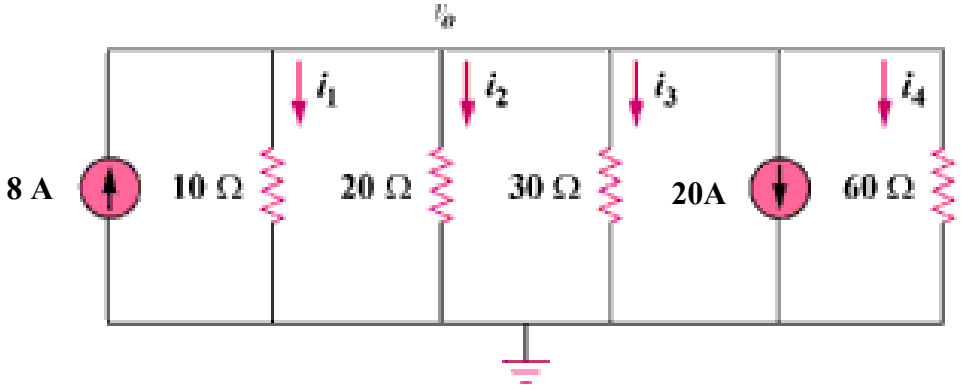


Figure 3.52
For Prob. 3.3.

4. Chapter 3, Problem 4

Given the circuit in Fig. 3.53, calculate the currents i_1 through i_4 using nodal analysis.

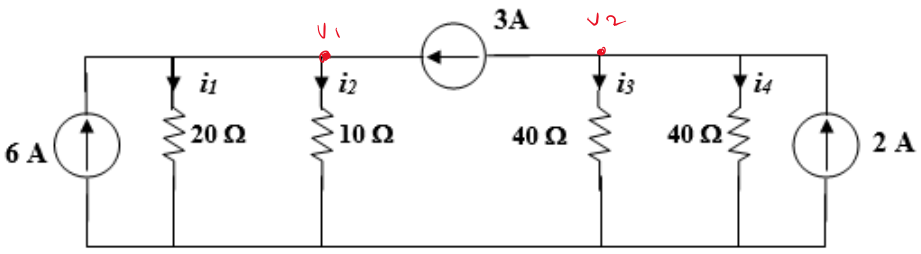


Figure 3.53
For Prob. 3.4.

5. Chapter 3, Problem 5.

Obtain v_0 in the circuit of Fig. 3.54 using Mesh analysis.

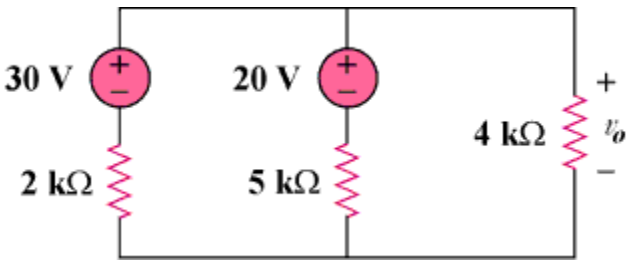


Figure 3.54
For Prob. 3.5.

6. Chapter 3, Problem 6.

Solve for V_1, in Fig. 3.55, using Mesh analysis.

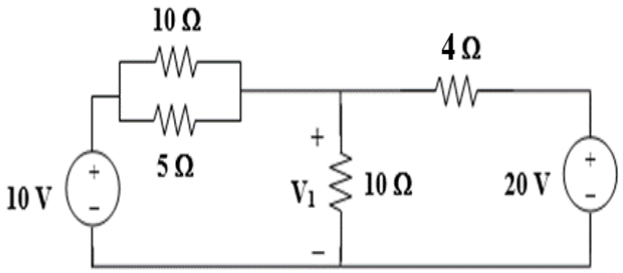


Figure 3.55
For Prob. 3.6.

7. Chapter 3, Problem 37.

Using mesh analysis, find v_0 in the circuit in Fig. 3.57.

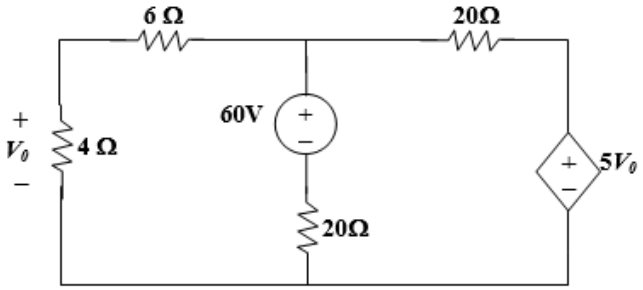


Figure
For Prob. 3.37.

8. Chapter 3, Problem 41.

Apply mesh analysis to find ***i*** in Fig. 3.87.

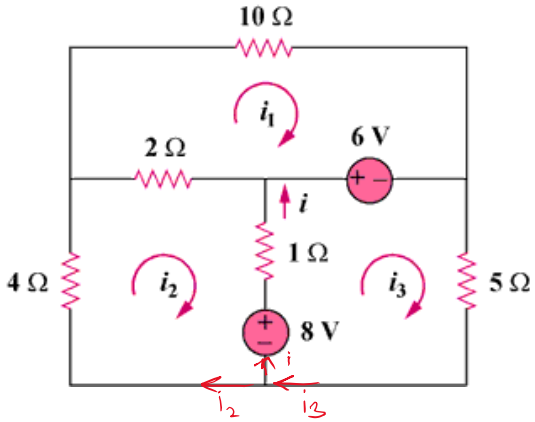


Figure 3.87

For Prob. 3.41.

9. Chapter 3, Problem 54

Given the circuit Fig. A, use mesh analysis to find the mesh currents.

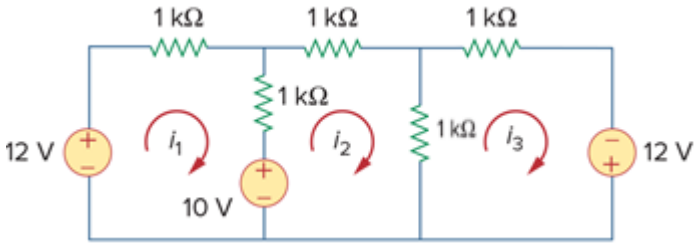
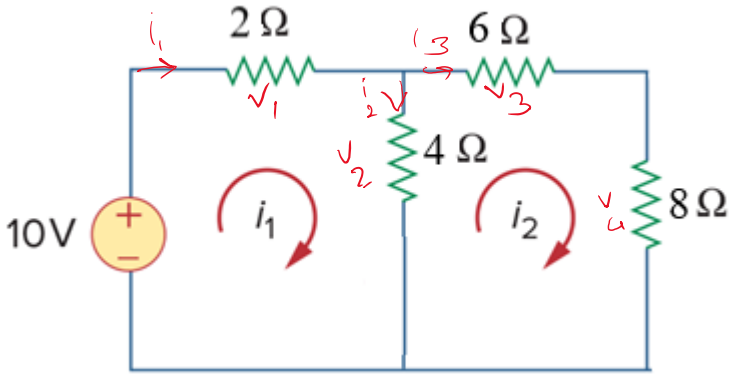


Figure 3.99
For Prob. 3.54

10. Chapter 3, Problem x

Given the circuit below, use mesh analysis to find the mesh currents voltages v_1, v_2 and v_3.



Reference: Fundamentals of Electric Circuits 7th edition, Charles Alexander and Matthew Sadiku, McGraw-Hill